

MuGate

Full-Project Soft / Application Relational Schema

Capstone-v2 · Declared FKs + application soft keys · PostgreSQL (Supabase)

Legend

- Solid line = declared PostgreSQL FOREIGN KEY
- - - - - Dashed line (soft) = application / soft key — column exists, NO DB FK

Soft relationships included (6)

- **CapstonePartners.userId -> Users.id**
JWT userId stored as text; ownership checks in CapstoneRepository
- **InternshipReviews.userId -> Users.id**
JWT userId stored as text; review CRUD scoped by userId
- **UserRoadmap.userId -> Users.id**
JWT userId stored as text; RoadMapRepository filters by userId
- **Admins.universityId -> Users.universityId**
Admin middleware + auth.routes LEFT JOIN on universityId
- **ScrapeQueue.runId -> ScraperRuns.id**
KnowledgeRepository.enqueueUrls tags queue rows with run UUID
- **Events.createdBy -> Users.name**
Manual events store display name (not id); denormalized attribution

Document pages

1. Page 1 — Full Schema Overview (FK + Soft)
2. Page 2 — Auth / Users (+ Admins soft)
3. Page 3 — Academic (+ UserRoadmap soft)
4. Page 4 — MuChat
5. Page 5 — Knowledge / Scraper (+ runId soft)
6. Page 6 — Capstone, Internships & Events

Truly unrelated after soft links

- ChatAnalytics - aggregate privacy-safe metrics only (no sessionId / userId columns)
- CapstoneIdeas - standalone idea catalog (no owner / user columns)

Why soft keys exist

- Legacy SQL Server ensureTables used NVARCHAR userId (text) without FK declarations.
- UUID user ids stored as text in feature tables → type mismatch blocks easy PG FKs.
- Admins keyed by portal universityId (natural key), not Users.id UUID.
- ScrapeQueue.runId left nullable / unconstrained so queue rows survive run cleanup.
- Events.createdBy is an audit display string (name), not a resolvable user UUID.

Derived from Capstone-v2 / MuGate PostgreSQL schema (supabase/migrations/20260714120000_mugate_schema.sql) plus verified soft keys from backend repositories/services/routes. Strict FK-only companion: MuGate-Full-Project-Relational-Schema.pdf

Not drawn as soft FKs: AcademicHistory.courseCode / UserRoadmap.courseCode -> Courses (shared natural key text, but no app join). Events.sourceId is external scraper id.